

Using Mathematica with Orgmode

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Background

I have been wanting to find a workflow which allows me to bypass writing a lot of TeX by hand for a while now. To that end I looked into using a computer algebra system (CAS). Naturally, my first choice was the FOSS [Maxima](#) (also because it uses Lisp under the hood). However, for all the reasons [listed here](#), relating to its accuracy, which have not been fixed even though the post was over 5 years ago, I ended up having to go with the closed source [Mathematica](#).

Packages

Support for Mathematica in modern orgmode is mainly through the use of [ob-mathematica](#), which is the official org-babel extension (from [contrib](#)) for working with Mathematica. However, [ob-mathematica](#) relies on the now-defunct [mma](#) package for font-locking, which is less than ideal. Thankfully, there exists the excellent [wolfram-mode](#) package which happens to be in MELPA as well. Finally, since the default return type of a [mathematica](#) block is an input-string meant to be used in another [mathematica](#) block, which is not useful when we work with [org-babel](#), we will use the excellent [mash.pl](#) utility [from here](#), as suggested by the [ob-mathematica](#) package to sanitize our output and set a unifying path.

So to recap, use your favorite manager to get:

- [ob-mathematica](#) (in contrib)
- [wolfram-mode](#) (MELPA)
- [mash.pl](#) ([from here](#))^[^fn:1]

After obtaining the packages, the configuration is then simply^[^fn:2]:

```
;; Load mathematica from contrib
(org-babel-do-load-languages 'org-babel-load-languages
                             (append org-babel-load-languages
                                     '((mathematica . t))
                                     ))

;; Sanitize output and deal with paths
(setq org-babel-mathematica-command "~/.local/bin/mash")

;; Font-locking
(add-to-list 'org-src-lang-modes '("mathematica" . wolfram))

;; For wolfram-mode
(setq mathematica-command-line "~/.local/bin/mash")
```

Results

LaTeX

Now we are in a position to simply evaluate content with font-locking. We will test our set up with an example lifted from the [ob-mathematica](#) [source-code](#).

Table 1: A table

1 4
2 4
3 6
4 8
7 0

```
(1+Transpose@x) // TeXForm
```

Where our header-line (with `#+begin_src`) is:

```
mathematica :var x=example-table :results latex
```

Sanity Checks

We can also test the example from the [blog post](#) earlier to test basic mathematical sanity.

```
Limit[Log[b - a + I eta], eta -> 0, Direction -> -1, Assumptions -> {a > 0, b > 0, a > b}]  
TeXForm[Limit[Log[b - a + I eta], eta -> 0, Direction -> 1, Assumptions -> {a > 0, b > 0, a > b}]]
```

$$(I \sqrt{\pi} + \operatorname{Log}[a - b]) \sqrt{\log(a - b)} - i\pi$$

Inline Math

Note that we can now also write fractions, integrals and other cumbersome TeX objects a lot faster with this syntax, like $\frac{x^3}{3}$. Where we are using the following snippet:

```
src_mathematica[:exports none :results raw]{Integrate[x^2,x] // TeXForm}
```

Plots

For plots, the standard `orgmode` rules apply, that is, we have to export to a file and return the name through our code snippet. Consider:

```
p=Plot[Sin[x], {x, 0, 6 Pi}, Frame->True];  
Export["images/sine.png",p];  
Print["images/sine.png"]
```

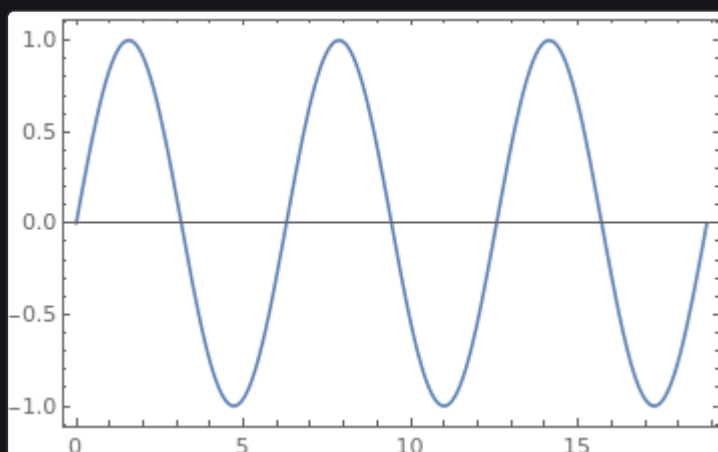


Figure 1: An exported Mathematica image

Where we have used `mathematica :results file` as our header line.

